

# Arranging circles of radii $1, 2, \dots, n$ around a central circle: a Supnick TSP and certified finite optima

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Corrective v2

**Correction of arXiv:2607.28654**

This corrective version preserves the original finite contribution and corrects its superseded asymptotic conjectures. The standalone sequel is available as arXiv:2609.13630 [7].

## Abstract

We study a discrete-geometric optimization problem: circles of radii  $1, 2, \dots, n$  are all externally tangent to a central circle, and the central radius  $R$  is minimized over cyclic orders of the surrounding circles. We prove that the chain-ordering component is governed by a fixed Supnick/anti-Monge traveling-salesman order. For every  $R$ , the angular-separation matrix is symmetric anti-Monge, so Supnick’s theorem gives one minimizing cyclic order, independent of  $R$ . This proves the conjectured “pyramid” order optimal whenever the corresponding chain necklace is geometrically realizable, and gives an unconditional lower bound in all cases.

Full geometric feasibility can fail because non-adjacent circle constraints are not captured by the chain equation; from  $n = 8$  the smallest circle can become a floating circle tangent only to the central circle. We formulate the full problem as a circular system of pairwise angular constraints, equivalently a simple temporal network, and certify global optima for  $3 \leq n \leq 14$  using branch-and-bound plus an independent 50-digit verifier. Continuation of the floating-circle pattern beyond that range remains conjectural. The v1 conjecture  $R^*(n) = n^2/8(1 + o(1))$  has been disproved by subsequent work [7]; this correction does not invalidate the finite results. The repository contains the saved certificate artifacts, verifier, and reproducibility commands.

## 1 Introduction

The following question was asked by the user Dan on Mathematics Stack Exchange in January 2023 [1] and appears as question 295 in the UQ benchmark of unsolved Stack Exchange questions [2].

Non-overlapping circles of radius  $1, 2, 3, \dots, n$  are all externally tangent to a middle circle. How should we arrange the surrounding circles in order to minimize the middle circle’s radius  $R$ ?

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Dan conjectured a specific “pyramid-fashion” arrangement and observed two phenomena that make the problem subtle: (i) the natural closure equation

$$\sum_{k=1}^n \arccos\left(1 - \frac{2r_k r_{k+1}}{(R + r_k)(R + r_{k+1})}\right) = 2\pi \quad (r_{n+1} = r_1) \quad (1)$$

treats consecutive circles as mutually tangent, yet (ii) the reported finite optima from  $n = 8$  admit circles tangent only to the central circle. This is an existence statement for the reported optima, not a conclusion about every optimal configuration or every larger  $n$ . A comment by Rei Henigman [1] verified the conjecture for  $n = 10$  by enumerating permutations of  $\{2, \dots, 10\}$ , manually excluding circle 1 from the chain.

This paper addresses the problem on three levels, keeping the epistemic status of each result explicit.

**1. The ordering problem is a Supnick TSP.** Writing  $\theta_R(a, b)$  for the minimum angular separation, at central radius  $R$ , between the centers of tangent circles of radii  $a$  and  $b$  (Section 2), we show that the matrix  $(\theta_R(r_i, r_j))_{i,j}$  is symmetric *anti-Monge* for every  $R > 0$  (Lemma 5). Minimizing total angle over cyclic orders is therefore a *maximum* traveling salesman problem on a Supnick matrix, which by a classical theorem of Supnick [3] (see also the survey [4]) is solved by one fixed tour independent of the matrix entries:

$$\sigma^* = \langle 1, n-1, 3, n-3, 5, n-5, \dots, n-2, 2, n \rangle.$$

A short self-consistency argument (the same order is optimal at *every*  $R$ ) transfers optimality from fixed  $R$  to the variable- $R$  problem (Theorem 6). The order  $\sigma^*$  coincides with Dan’s conjectured pyramid arrangement. In particular  $R_{\text{chain}}(\sigma^*)$ , defined by (1), is an unconditional lower bound for  $R^*(n)$ , and it is the exact optimum whenever the  $\sigma^*$ -necklace is geometrically realizable.

**2. Realizability breaks in a finite certified range.** The closure equation (1) ignores non-adjacent pairs. For  $n \geq 8$  the configuration it describes for  $\sigma^*$  is *not realizable*: circle 1, seated between circles  $n$  and  $n-1$ , no longer keeps them apart, i.e.  $\theta_R(n, 1) + \theta_R(1, n-1) < \theta_R(n, n-1)$ . The reported certified optima for  $8 \leq n \leq 14$  admit circle 1 as a *floating* circle, tangent to the central circle only. This finite statement does not follow from the seam obstruction alone. We model the full problem as a circular system of  $\binom{n}{2}$  pairwise angular constraints — equivalently a simple temporal network (STN) — and give a certified algorithm for it (Section 4). Exhaustive certified optimization for  $3 \leq n \leq 14$  (Section 5) reveals the reported finite regimes below, described by two computable quantities of the Supnick tour on  $\{2, \dots, n\}$ : its geometric validity and the largest Descartes pocket it opens.

| regime                 | $n$               | reported floating set | reported chain                        |
|------------------------|-------------------|-----------------------|---------------------------------------|
| I (full necklace)      | $3 \leq n \leq 7$ | $\emptyset$           | $\sigma^*$ on $\{1, \dots, n\}$       |
| II (paid floater)      | $n = 8, 9$        | $\{1\}$               | distorted tour on $\{2, \dots, n\}$   |
| III (free floater)     | $n = 10, 11, 12$  | $\{1\}$               | $\sigma^*$ on $\{2, \dots, n\}$       |
| IV (second breakdown)  | $n = 13$          | $\{1\}$               | constrained tour on $\{2, \dots, n\}$ |
| II' (cascade, level 2) | $n = 14$          | $\{1, 2\}$            | constrained tour on $\{3, \dots, n\}$ |

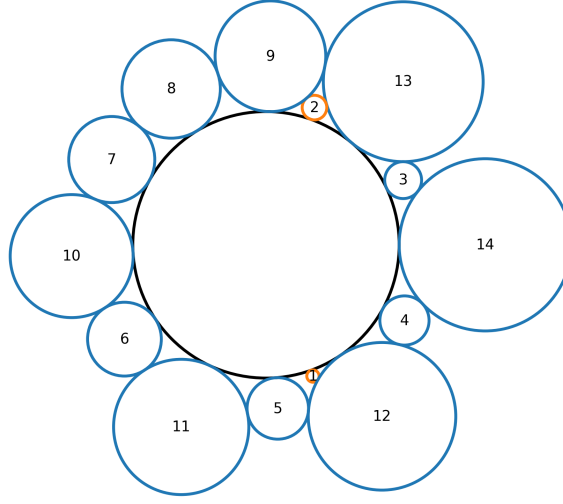


Figure 1: A reported certified optimum for  $n = 14$ ; circles 1, 2 admit strictly floating placements at the reported radius and are highlighted. A plotted witness need not itself display every available strict slack.

**3. Quantitative results.** We report certified optima for  $3 \leq n \leq 14$ , with their high-precision reconstructions and reported binding/floating structure (Table 1); the chain-radius-maximizing arrangement, which is the *minimum*-tour counterpart  $\langle 1, 3, 5, \dots, n, \dots, 4, 2 \rangle$  of Supnick’s theorem (Proposition 8); and the correction of the disproved v1 asymptotic conjecture (Section 6). Global certification retains the  $10^{-10}$  radius guard; 50-digit arithmetic does not mean 50-digit global certification.

**Epistemic ledger.** The evidence classes are explicit. *Proved:* Lemmas 1 and 5, Theorem 6, and the chain-level statement of Proposition 8. *Computer-certified finite results:* Table 1, supported by the exhaustive search record and the numerical guards in Section 4. The pocket values, finite realizability diagnostics and complementary-tour values are *numerical fixed-order evidence*; their printed precision does not expand the global certificate. *Heuristic or unresolved:* Table 2 and the global questions in Remark 13. The earlier coefficient and deficit conjectures are *disproved*, as recorded in Remark 14; subsequent exact fixed-order results are cited separately.

**Related work.** Supnick’s fixed-tour theorem [3] and the Monge/Supnick taxonomy of polynomially solvable TSPs are surveyed in [4]. Circle packing literature includes container problems (packing circles *inside* regions, see e.g. [6]); the present paper concerns unequal circles externally tangent to a central one. The tangency mechanics use Descartes’ Circle Theorem [5]. The asymptotic sequel [7] discusses the related line/shelf model. No exhaustive novelty or priority claim is required here.

## 2 The model

Let the central circle have radius  $R > 0$  and center  $O$ . A surrounding circle of radius  $r$  tangent externally to it has its center at distance  $R + r$  from  $O$ ; its position is a single angle  $\varphi$ . The whole configuration is a vector of angles.

**Lemma 1** (Exact angular reformulation). *Circles of radii  $a, b$ , both externally tangent to the central circle, with angular positions separated by  $\Delta \in [0, \pi]$ , are non-overlapping if and only if*

$$\Delta \geq \theta_R(a, b) := 2 \arcsin \sqrt{\frac{ab}{(R+a)(R+b)}}.$$

Moreover  $\theta_R(a, b) \in (0, \pi)$  always,  $\theta_R$  is symmetric, strictly decreasing in  $R$ , and strictly increasing in each radius.

*Proof.* By the law of cosines the squared distance between the centers is  $(R+a)^2 + (R+b)^2 - 2(R+a)(R+b) \cos \Delta$ , and non-overlap means this is at least  $(a+b)^2$ . Rearranging gives  $\cos \Delta \leq 1 - \frac{2ab}{(R+a)(R+b)}$ , i.e.  $\Delta \geq \theta_R(a, b)$  with  $\theta_R$  as stated, using  $\arccos(1-2s^2) = 2 \arcsin s$  for  $s \in [0, 1]$ . The argument of the square root is strictly less than 1 since  $(R+a)(R+b) - ab = R(R+a+b) > 0$ , whence  $\theta_R < \pi$ ; the monotonicity claims are immediate from the formula.  $\square$

Thus the problem is: minimize  $R$  such that there exist angles  $\varphi_1, \dots, \varphi_n$  with circular separations  $\geq \theta_R(r_i, r_j)$  for *all* pairs. For a fixed cyclic order this is a circular system of difference constraints — a simple temporal network — and feasibility at fixed  $R$  is decidable by shortest paths (Section 4). Two derived quantities recur:

**Definition 2.** For a cyclic order  $\sigma$  of at least three positive radii,  $R_{\text{chain}}(\sigma)$  is the unique solution of  $\sum_i \theta_R(\sigma_i, \sigma_{i+1}) = 2\pi$ . (Existence and uniqueness: the left side tends to  $|\sigma| \pi > 2\pi$  as  $R \rightarrow 0^+$ , to 0 as  $R \rightarrow \infty$ , and is strictly decreasing.) For a feasible pair (order,  $R$ ), summing the adjacent gaps shows  $\sum_i \theta_R(\sigma_i, \sigma_{i+1}) \leq 2\pi$  and hence  $R \geq R_{\text{chain}}(\sigma)$ : the chain value is an unconditional per-order lower bound.

**Definition 3** (Floating circle). A circle  $c$  is *floating* at a global optimum if there exists an optimal configuration in which  $c$  is tangent only to the central circle (every pairwise constraint involving  $c$  is strictly slack).

**Definition 4** (Descartes pocket). For adjacent, mutually tangent circles  $a, b$  on the necklace at central radius  $R$ , the *pocket* is the largest circle externally tangent to the central circle and to both: by Descartes' Circle Theorem its curvature is  $k_{\text{pocket}} = \frac{1}{R} + \frac{1}{a} + \frac{1}{b} + 2\sqrt{\frac{1}{Ra} + \frac{1}{ab} + \frac{1}{Rb}}$ . A circle of radius  $\rho$  fits in this pocket without overlap with its two boundary circles iff  $\rho \leq 1/k_{\text{pocket}}$ , equivalently  $\theta_R(a, \rho) + \theta_R(\rho, b) \leq \theta_R(a, b)$ . Equality makes it tangent to both boundaries; strict inequality permits slack against those two circles. Strict floating in the full configuration additionally requires every other pair constraint involving that circle to be slack.

## 3 The optimal cyclic order is a fixed Supnick tour

**Lemma 5** (Supermodularity). *For every  $R > 0$  the matrix  $(\theta_R(r_i, r_j))$ , rows and columns indexed by the radii in increasing order, is symmetric strictly anti-Monge: for  $a < a'$  and  $b < b'$ ,*

$$\theta_R(a, b) + \theta_R(a', b') > \theta_R(a, b') + \theta_R(a', b).$$

*Proof.* Write  $\theta_R(a, b) = 2 \arcsin(t_a t_b)$  with  $t_x = \sqrt{x/(R+x)} \in (0, 1)$ , which is strictly increasing in  $x$ . For  $f(x, y) = \arcsin(xy)$  on  $(0, 1)^2$ ,

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y} \frac{y}{\sqrt{1-x^2 y^2}} = \frac{(1-x^2 y^2) + x^2 y^2}{(1-x^2 y^2)^{3/2}} = (1-x^2 y^2)^{-3/2} > 0,$$

so  $f$  is strictly supermodular; supermodularity is preserved by strictly increasing reparametrizations of each coordinate, and the claim follows.  $\square$

Equivalently,  $-\theta_R$  is a Supnick matrix — symmetric Monge off the diagonal, diagonal entries being irrelevant for tours (cf. [4, Prop. 2.13]). Supnick’s theorem [3] covers both extreme tours; in the form stated in [4] (Theorem 2.5 for the minimum, and the max-TSP discussion following Theorem 2.12, both attributed there to [3]), the minimum TSP tour on Supnick matrices is  $\langle 1, 3, 5, \dots, n, \dots, 6, 4, 2 \rangle$  and the *maximum* TSP tour is

$$\sigma^* = \langle 1, n-1, 3, n-3, 5, n-5, \dots, n-2, 2, n \rangle,$$

*independently of the matrix entries.* Maximizing  $\sum(-\theta_R)$  is minimizing  $\sum \theta_R$ , hence:

**Theorem 6** (Optimal order). *Let  $n \geq 3$  and let  $r_1 < \dots < r_n$  be arbitrary positive radii.*

- (i) *For every  $R > 0$ , among all cyclic orders the tour  $\sigma^*$  minimizes  $\sum_i \theta_R(r_{\sigma(i)}, r_{\sigma(i+1)})$ .*
- (ii)  *$\sigma^*$  minimizes  $R_{\text{chain}}$  over all cyclic orders; hence  $R^* \geq R_{\text{chain}}(\sigma^*)$  unconditionally.*
- (iii) *If the  $\sigma^*$ -necklace at  $R_{\text{chain}}(\sigma^*)$  is geometrically realizable (no non-adjacent pair violated), then it is a global optimum:  $R^* = R_{\text{chain}}(\sigma^*)$ .*

*Proof.* (i) is Lemma 5 plus Supnick’s theorem. (ii): fix any order  $\sigma$  and let  $R = R_{\text{chain}}(\sigma)$ . By (i),  $\sum \theta_R(\sigma^*) \leq \sum \theta_R(\sigma) = 2\pi$ ; since  $\sum \theta_R(\sigma^*)$  is strictly decreasing in  $R$  and equals  $2\pi$  at  $R_{\text{chain}}(\sigma^*)$ , we get  $R_{\text{chain}}(\sigma^*) \leq R$ . The unconditional bound follows since every feasible  $(\sigma, R)$  has  $R \geq R_{\text{chain}}(\sigma)$ . (iii): realizability makes the lower bound attained.  $\square$

*Remark 7.* Theorem 6 holds for *arbitrary* distinct radii, not only  $1, \dots, n$ ; the arithmetic sequence enters only in the realizability analysis below. For  $n = 10$  the lower bound of (ii) is  $\approx 9.90$ , the value observed in [1] when circle 1 is (incorrectly) kept in the chain; the true optimum is  $9.9799\dots$

**Proposition 8** (Maximum chain radius). *For every  $R$  the cyclic order maximizing  $\sum_i \theta_R$  is the complementary Supnick tour  $\sigma_{\dagger} = \langle 1, 3, 5, \dots, n, \dots, 6, 4, 2 \rangle$ ; consequently  $\sigma_{\dagger}$  maximizes  $R_{\text{chain}}$ . For  $3 \leq n \leq 13$ , the reported fixed-order checks find its necklace realizable; Table 3 records the corresponding numerical values. Thus it supplies the reported maximum among realizable adjacent-chain necklaces in that finite range. This does not assert that  $\sigma_{\dagger}$  maximizes  $R_{\text{full}}$  over all cyclic orders.*

## 4 Exact algorithm and certification

**Feasibility oracle.** Fix a cyclic order and  $R$ . Unrolling the circle at one circle ( $\varphi_0 = 0$ ) gives, for every pair  $i < j$  of positions, the two constraints  $\varphi_j - \varphi_i \geq \theta_R(r_i, r_j)$  and  $\varphi_j - \varphi_i \leq 2\pi - \theta_R(r_i, r_j)$ . This is a simple temporal network; feasibility is the absence of a negative cycle

in the associated distance graph and is decided by Floyd–Warshall in  $O(n^3)$ . The minimum feasible  $R$  for the order,  $R_{\text{full}}(\sigma)$ , is found by bisection (the constraint set relaxes monotonically in  $R$ ), bracketed below by  $R_{\text{chain}}(\sigma)$ . A witness configuration is recovered from shortest-path potentials and verified independently in Cartesian coordinates.

**Lower bounds and search.** For the search over orders we use

$$\text{LB}(\sigma) = \max \{R_{\text{chain}}(\sigma), R_{\text{chain}}(\sigma \setminus \{1\}), R_{\text{chain}}(\sigma \setminus \{1, 2\})\},$$

where  $\sigma \setminus S$  denotes the induced cyclic order, and terms with fewer than three surviving radii are omitted. Each retained term is valid because every induced adjacency is one of the full pairwise constraints on the original radii. Orders are enumerated canonically (largest radius pinned, reflections halved), LB is computed vectorized for all  $(n-1)!/2$  orders, and  $R_{\text{full}}$  is evaluated in ascending LB until the next bound meets the incumbent: this certifies global optimality up to an absolute guard of  $10^{-10}$  in  $R$ . The historical error budget and calibration supporting this numerical guard are: bisections run to relative tolerance  $10^{-13}$ ; one bound evaluation is an  $n$ -term sum of arcsin values, with relative error  $O(n \varepsilon_{64})$ ; and the float64 bounds were calibrated against 50-digit recomputation on random samples of orders, with maximal observed deviation  $1.8 \times 10^{-14}$  (over  $10^5$  random orders for each  $8 \leq n \leq 14$ ; maximal overestimate  $1.5 \times 10^{-14}$ ), more than three orders of magnitude below the guard. These numerical certificates retain the original floating-point error guards: saved-frontier verification checks retained orders and coverage summaries, not every excluded order with directed interval arithmetic. Independently of the search code, the repository ships a standalone verifier (`verify.py`) that re-derives each claimed optimum and re-checks every order on the pruning frontier (between 1 and 11 orders per  $n$ ) in 50-digit arithmetic. Optimal binding structures were re-verified in 50-digit arithmetic (`mpmath`); the reported high-precision incumbent reconstructions have essential residual slack below  $10^{-40}$ . Extra printed digits describe those reconstructions; the global optimum guarantee remains the stated  $10^{-10}$  tolerance. The  $n = 13$  certification enumerates  $12!/2 \approx 2.4 \times 10^8$  orders ( $\approx 1.8$  h on a consumer 8-core laptop);  $n = 14$  enumerates  $13!/2 \approx 3.1 \times 10^9$  orders ( $\approx 22$  h). Code and artifacts: <https://github.com/falker47/ringmin>.

**Essential binding structure.** At the reported fixed order and radius, the verifier calls a pairwise constraint *essential* when increasing its required separation by  $10^{-9}$  makes the STN infeasible. To verify that a circle admits a strictly floating placement, it separately increases *all* pairwise separation requirements involving that circle by  $10^{-9}$  and checks feasibility. The absence of reported essential pairs is a consistency check, not a substitute for this joint slack test. These are existence statements at the recorded numerical tolerances, not assertions that a circle is slack in every optimal placement or in the plotted witness.

**Reproducing the saved certificates.** From the repository root, restore the byte-preserved historical progress logs and run:

```
python scripts/frontier_logs.py restore
python verify.py --start 3 --stop 14
```

The option `-skip-frontier` is only a smoke check. Rechecking the saved evidence does not rerun exhaustive search generation, and local success does not imply hosted CI or external acceptance.

## 5 Certified optima for $3 \leq n \leq 14$ and finite regimes

Table 1: Reported certified optima with global radius guard  $10^{-10}$ ; incumbent reconstructions are rounded to 12 decimals (the unchanged 30-digit table is reproduced in the Appendix). Each cycle is one reported optimal order. The floating column records the existence of strict-slack placements; its displayed hosting gap is not unique. “Essential pair” identifies the reported non-adjacent boundary pair bridged by a listed floater.

| $n$ | $R^*(n)$                | optimal cycle                      | floating | essential pair   |
|-----|-------------------------|------------------------------------|----------|------------------|
| 3   | 0.260869565217 (= 6/23) | (3,1,2)                            | —        | —                |
| 4   | 0.844453589561          | (4,1,3,2)                          | —        | —                |
| 5   | 1.695494081203          | (5,1,4,3,2)                        | —        | —                |
| 6   | 2.794919518897          | (6,1,5,3,4,2)                      | —        | —                |
| 7   | 4.153189553744          | (7,1,6,3,4,5,2)                    | —        | —                |
| 8   | 5.767794284590          | (8,1,6,4,5,3,7,2)                  | {1}      | (8, 6)           |
| 9   | 7.726726552611          | (9,2,8,1,5,6,4,7,3)                | {1}      | (8, 5)           |
| 10  | 9.979907385863          | (10,2,9,4,7,1,6,5,8,3)             | {1}      | (7, 6)           |
| 11  | 12.488720487188         | (11,2,10,4,8,6,7,1,5,9,3)          | {1}      | (7, 5)           |
| 12  | 15.258870430448         | (12,2,11,4,9,6,7,8,5,1,10,3)       | {1}      | (5, 10)          |
| 13  | 18.317563047217         | (13,3,1,12,2,10,6,8,7,9,5,11,4)    | {1}      | (3, 12)          |
| 14  | 21.665395182215         | (14,3,13,2,9,8,7,10,6,11,5,1,12,4) | {1, 2}   | (13, 9), (5, 12) |

For  $3 \leq n \leq 7$  a reported optimal cycle is  $\sigma^*$  on  $\{1, \dots, n\}$  (regime I), as guaranteed by Theorem 6(iii) once realizability is checked. The transitions are governed by two computable properties of the Supnick tour on the reduced set  $\{2, \dots, n\}$ :

**Proposition 9** (First breakdown). *In the  $\sigma^*$ -necklace on  $\{1, \dots, n\}$ , circle 1 sits between  $n$  and  $n - 1$ . The realizability inequality  $\theta_R(n, 1) + \theta_R(1, n - 1) \geq \theta_R(n, n - 1)$  at  $R = R_{\text{chain}}(\sigma^*)$  holds for  $3 \leq n \leq 7$  and fails for  $8 \leq n \leq 13$  (violated pair  $(n, n - 1)$ ; at  $n = 13$  also  $(13, 11)$  across circle 2). Hence regime I ends at  $n = 7$ . The continuation to every  $n \geq 8$ , conjectural in v1, is now a fixed-order theorem [8]. It does not by itself classify floating circles in global optima.*

**Proposition 10** (Free vs. paid floater). *Let  $\rho_{\max}(n)$  be the largest Descartes pocket of the  $\sigma^*$ -necklace on  $\{2, \dots, n\}$  at its chain radius. The reported numerical values for  $n = 8, \dots, 12$  are*

$$\rho_{\max} \approx 0.858, 0.988, 1.139, 1.262, 1.409.$$

*For  $n = 10, 11, 12$  one has  $\rho_{\max} > 1$ , and the separate all-pairs checks admit a strictly floating placement of circle 1. The reported full radii agree with  $R_{\text{chain}}(\sigma^*$  on  $\{2, \dots, n\})$  within the stated certification tolerance (regime III). The pocket comparison alone tests only its two boundary circles. For  $n = 8, 9$ ,  $\rho_{\max} < 1$ : the reported optimum distorts the reduced tour to open a unit pocket, at extra cost  $\delta(8) \approx 0.04080$ ,  $\delta(9) \approx 0.00532$  over the (then unattainable) reduced chain bound (regime II), with the same numerical certificate scope.*

**Proposition 11** (Second breakdown). *At  $n = 13$  the Supnick tour on  $\{2, \dots, 13\}$  is itself unrealizable: circle 2, seated between 13 and 12, violates  $\theta_R(13, 2) + \theta_R(2, 12) = 0.80448 < 0.83488 = \theta_R(13, 12)$  at  $R = 18.3176$ . The reported certified optimum places circle 2 to the pair  $(12, 10)$ , where the separation constraint holds with slack  $6 \times 10^{-3}$  rad; circle 1 floats in the  $(3, 12)$  pocket (radius 1.086). Thus the mechanism that expelled circle 1 at  $n = 8$  reaches circle 2 at  $n = 13$  (regime IV).*

*Remark 12* (Structural summary). For each  $3 \leq n \leq 14$ , the reported optimum has a Supnick-like constrained backbone and the listed floating circles in available pockets. This describes the saved solutions rather than all optimal contact graphs. The observed finite pattern repeats at its next level: circle 2 first appears in the reported floating set at  $n = 14$  (Table 1) via a *paid* distortion that opens a pocket of radius 2.013 between circles 13 and 9 — the regime-II mechanism one level up, at extra cost 0.0565 over the (then unattainable) chain bound of the Supnick tour on  $\{3, \dots, 14\}$ . At  $n = 16$  the best known value coincides to  $10^{-10}$  with the chain value of the Supnick tour on  $\{3, \dots, 16\}$ , the level-2 analogue of regime III. Heuristically, circle 3 joins by  $n = 18$  (Table 2). Moreover the Supnick tour on  $\{3, \dots, n\}$  itself becomes unrealizable at  $n = 17$  — circle 3 failing at the (17, 16) seam with slack  $-0.0104$  — so the *seam-failure* onsets for circles 1, 2, 3 are  $n = 8, 13, 17$ : each *computed* cascade level reproduces the full regime pattern of the previous one. The appearances at  $n = 8, 14, 18$  for circles 1, 2, 3 concern these reported solutions, with  $n = 18$  heuristic. They do not determine the number of floating circles in every optimum. The global continuation questions are separated below from the now-proved fixed-order seam theory.

*Remark 13* (Status of the v1 cascade conjecture). The fixed-order part is now proved: for every fixed integer  $k \geq 1$  the canonical Supnick necklace on  $\{k, \dots, n\}$  eventually remains unrealizable at its chain root [8]. The first three exact seam thresholds are 8, 13, 17. This does not resolve the global part. Whether each fixed circle eventually admits a strictly floating placement in some global optimum, the stronger v1 assertion that it is slack in every optimal placement, and indefinite repetition of paid-then-free regimes remain unresolved. No one of these quantifiers is inferred from a fixed-order seam or a saved contact graph.

Table 2: Historical best-known arrangements for  $15 \leq n \leq 18$  (local search, 200–600 restarts per  $n$ ; *non-exhaustive*). They are heuristic feasible upper bounds, not certified global optima; floating labels concern the reported candidates. The last column counts restarts reaching the best value.

| $n$ | best $R$        | floating      | cycle                                          | hits |
|-----|-----------------|---------------|------------------------------------------------|------|
| 15  | 25.290151636714 | $\{1, 2\}$    | (15,3,14,5,12,2,9,1,8,10,7,11,6,13,4)          | 59   |
| 16  | 29.194988498141 | $\{1, 2\}$    | (16,1,4,14,6,12,8,10,9,2,11,7,13,5,15,3)       | 18   |
| 17  | 33.394425022062 | $\{1, 2\}$    | (17,4,16,3,15,6,1,13,8,11,10,9,2,12,7,14,5)    | 19   |
| 18  | 37.870755667393 | $\{1, 2, 3\}$ | (18,4,17,3,14,8,12,10,11,9,2,13,7,15,6,16,1,5) | 9    |

## 6 Correction of the v1 asymptotic conjectures

For  $R \gg n$  one has  $\theta_R(a, b) = 2\sqrt{ab}/R(1 + O(n/R))$ , so the closure equation gives  $R \approx \frac{1}{\pi} \sum_i \sqrt{r_i r_{i+1}}$ . Since  $\sqrt{ab}$  is supermodular, Theorem 6(i) applies verbatim: the minimizing order of  $\sum \sqrt{r_i r_{i+1}}$  is again  $\sigma^*$ , whose value on  $1, \dots, n$  is

$$\min_{\sigma} \sum_i \sqrt{r_{\sigma(i)} r_{\sigma(i+1)}} = (1 + o(1)) \int_0^n \sqrt{x(n-x)} dx = \frac{\pi n^2}{8} (1 + o(1)),$$

because adjacent pairs in  $\sigma^*$  are  $\{k, n - k + O(1)\}$  with each odd  $k$  contributing twice. This is the adjacent-chain relaxation; it gives no all-pairs-feasible construction at that coefficient.

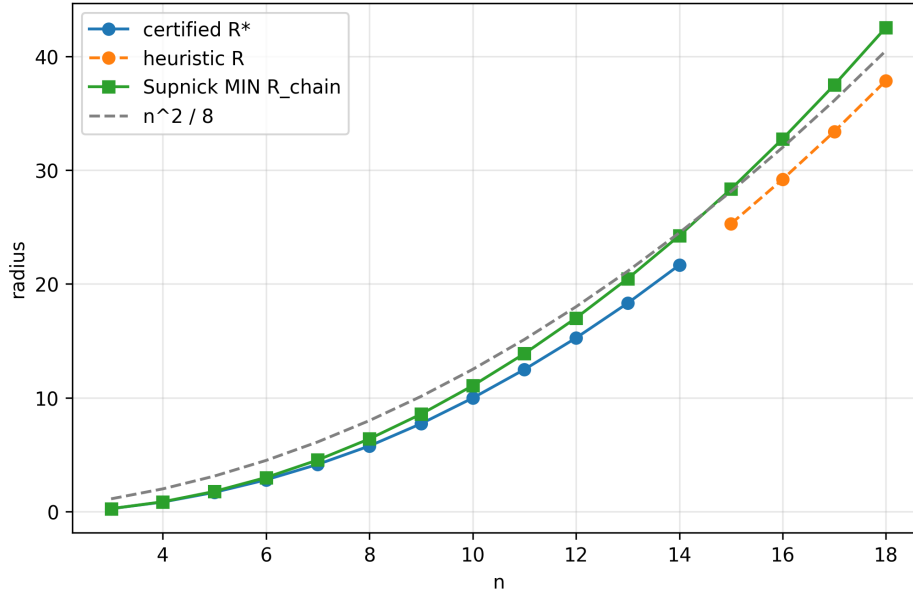


Figure 2: Unchanged historical finite comparison: reported minima (heuristic upper bounds beyond  $n = 14$ ), complementary-tour chain values, and the  $n^2/8$  reference curve. The latter is the *disproved v1 conjectural asymptote*, not the true global asymptote. The other comparison is a fixed-order chain curve, not a certificate for maximizing full geometric radii.

In particular, the v1 argument based on a small number of floating circles did not supply the missing feasibility transfer. The chain calculation therefore does not establish  $R^*(n) \sim n^2/8$ .

*Remark 14* (Disproved v1 asymptotic conjectures). Version 1 conjectured both  $R^*(n) = n^2/8(1+o(1))$  and the stronger deficit statement  $n^2/8 - R^*(n) = O(\sqrt{n})$ . Both are false: subsequent work proves a global asymptotic coefficient strictly larger than  $1/8$  [7]. The historical finite deficits 2.52, 2.64, 2.74, 2.81, 2.83 for  $n = 10, \dots, 14$  and 2.83, 2.81, 2.73, 2.63 for the heuristic  $n = 15, \dots, 18$  remain numerical observations only. The heuristic deficits are lower bounds on the true deficits when their candidate radii are feasible upper bounds. These finitely many values imply no asymptotic rate.

The standalone sequel [7] proves  $R^*(n) = C_*n^2 + o(n^2)$  with  $C_* > 1/8$  and gives an effective finite-word variational characterization. Its all-pairs reduction, bounds and recovery proofs belong to that separate work. This correction imports their consequence for the former conjectures without reproducing the new theory, and leaves the finite certificates unchanged.

## 7 Open problems

(1) Determine global finite optima beyond  $n = 14$ ; the now-proved fixed-order seam classification [8] does not settle them. (2) Characterize floating circles with explicit quantifiers over optimal placements: distinguish existence of a strict-slack placement from slack in every optimum, and do not infer a growth rate from the finite appearances in Table 1. The global continuation questions in Remark 13 remain open. (3) The same machinery applies to radii  $r_k = k^\alpha$  or arbitrary sequences: Theorem 6 is general, the cascade analysis is not. (4) Three dimensions:

spheres of radii  $1, \dots, n$  tangent to a central sphere — the ordering structure disappears; what replaces Supnick?

## Acknowledgments

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<https://github.com/falker47/ringmin/tree/3beb8d70c5b3748d370a92855847bdf574e5a14f>.

## A High-precision values and figures

Table 3: The complementary Supnick tour  $\sigma_{\dagger}$  (Proposition 8): reported numerical chain values and finite necklace realizability. These values do not certify a maximum of  $R_{\text{full}}$  over all orders.

| $n$           | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10      | 11      | 12      | 13      |
|---------------|--------|--------|--------|--------|--------|--------|--------|---------|---------|---------|---------|
| $R_{\dagger}$ | 0.2609 | 0.8673 | 1.7786 | 3.0001 | 4.5375 | 6.3934 | 8.5690 | 11.0647 | 13.8806 | 17.0164 | 20.4720 |

| $n$ | $R^*$                            |
|-----|----------------------------------|
| 3   | 0.260869565217391304347826086957 |
| 4   | 0.844453589560855604347528524674 |
| 5   | 1.69549408120271081351328017371  |
| 6   | 2.79491951889692485617024406797  |
| 7   | 4.15318955374381246513863858202  |
| 8   | 5.7677942845896143026361805725   |
| 9   | 7.72672655261128921886246177604  |
| 10  | 9.97990738586347760966552641468  |
| 11  | 12.4887204871876588517468786264  |
| 12  | 15.2588704304484933617250043503  |
| 13  | 18.3175630472173206282821941532  |
| 14  | 21.6653951822145150956462891793  |

Table 4: Certified optima, 30 significant digits.

The following tables reproduce the original appendix data unchanged. Their extra digits report high-precision fixed-order reconstructions, not a global error smaller than  $10^{-10}$ . The historical label “Worst-arrangement comparison” refers to the complementary Supnick chain tour; the  $n > 14$  rows are fixed-order comparisons and are not additional global optimum certificates.

Per- $n$  configuration figures for all  $3 \leq n \leq 18$  are available in the repository.

| $n$ Supnick min tour                                               | $R_{\text{chain}}$ | $R_{\text{full}}$ |
|--------------------------------------------------------------------|--------------------|-------------------|
| 3 [1, 3, 2]                                                        | 0.260869565217     | 0.260869565217    |
| 4 [1, 3, 4, 2]                                                     | 0.867281640255     | 0.867281640255    |
| 5 [1, 3, 5, 4, 2]                                                  | 1.778550949884     | 1.778550949884    |
| 6 [1, 3, 5, 6, 4, 2]                                               | 3.000138442323     | 3.000138442323    |
| 7 [1, 3, 5, 7, 6, 4, 2]                                            | 4.537467833562     | 4.537467833562    |
| 8 [1, 3, 5, 7, 8, 6, 4, 2]                                         | 6.393358320114     | 6.393358320114    |
| 9 [1, 3, 5, 7, 9, 8, 6, 4, 2]                                      | 8.568988935955     | 8.568988935955    |
| 10 [1, 3, 5, 7, 9, 10, 8, 6, 4, 2]                                 | 11.064726494858    | 11.064726494858   |
| 11 [1, 3, 5, 7, 9, 11, 10, 8, 6, 4, 2]                             | 13.880577932626    | 13.880577932626   |
| 12 [1, 3, 5, 7, 9, 11, 12, 10, 8, 6, 4, 2]                         | 17.016408238128    | 17.016408238128   |
| 13 [1, 3, 5, 7, 9, 11, 13, 12, 10, 8, 6, 4, 2]                     | 20.472039120681    | 20.472039120681   |
| 14 [1, 3, 5, 7, 9, 11, 13, 14, 12, 10, 8, 6, 4, 2]                 | 24.247291190092    | 24.247291190092   |
| 15 [1, 3, 5, 7, 9, 11, 13, 15, 14, 12, 10, 8, 6, 4, 2]             | 28.342000288518    | 28.342000288518   |
| 16 [1, 3, 5, 7, 9, 11, 13, 15, 16, 14, 12, 10, 8, 6, 4, 2]         | 32.756022334220    | 32.756022334220   |
| 17 [1, 3, 5, 7, 9, 11, 13, 15, 17, 16, 14, 12, 10, 8, 6, 4, 2]     | 37.489233291550    | 37.489233291550   |
| 18 [1, 3, 5, 7, 9, 11, 13, 15, 17, 18, 16, 14, 12, 10, 8, 6, 4, 2] | 42.541527273519    | 42.541527273519   |

Table 5: Worst-arrangement comparison; Supnick min tour values.